KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2017
Department of Computer Science and Engineering
CSE 4211
Machine learning

TIME: 3 hours FULL MARKS: 210

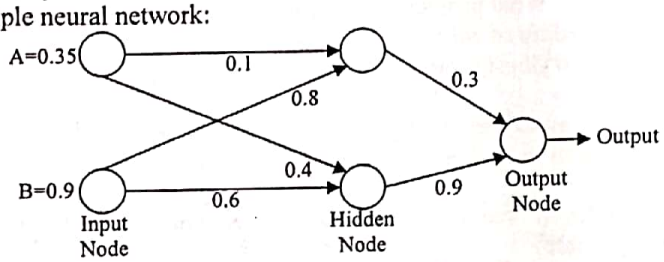
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Machine Learning? Briefly discuss some practical applications of Machine Learning. (08)
   b) Distinguish between supervised and unsupervised learning with example. (07)
   c) How do you differentiate regression and classification problems in Machine Learning? Give some practical examples. (10)
   d) Briefly explain the curse of dimensionality problem. (10)

2. a) What do you mean by feed forward network? (05)
   b) Consider the simple neural network: (10)



   (i) Perform a forward pass on the network.
   (ii) Perform a backward pass on the network
   (iii) Perform a further forward pass and comment on the result.
   Use sigmoid function as activation function.
   c) Explain in difficulties to solve XOR problem with single neuron. Solve XOR problem through manipulation by hidden neurons for AND and OR logic. (10)
   d) Draw minimal neural models for the following two logics having inputs A, B and C. (10)
   (i) NN1=(A+B)C' (ii) NN2=ABC'

3. a) What is the effect of learning rate in BP? How momentum parameters speed up training in BP? (10)
   b) What are the basic differences between RBF network and MLP. (08)
   c) What is local gradient in BP training procedure? For a neural network with two hidden layers, write local gradients for each layer. (10)
   d) Differentiate learning and generalization? Which one is more desirable and why? (07)

4. a) What is the purpose of mathematical optimization? Define following terminologies with respect to mathematical optimization: (i) Objective function (ii) variables/parameter (iii) Constraints (iv) Constrained and unconstrained optimization (v) language multiplier. (07)
   b) How to solve an unconstrained optimization problem using Gradient descent. Explain in detail. (08)
   c) Solve the following constrained optimization problems: (i) Given the objective function (20)

$f(x,y) = x^3 + y^3$ *subject* to

$g_1(x,y) = x^2 - 1 \geq 0$

$g_2(x,y) = y^2 - 1 \geq 0$

Find the optimum value.

(ii) A company has determined that its production is the cobb-douglas function, $f(x,y) = x^{2/3}y^{1/3}$ where x=number of labour hours, y=number of capital units, the budget constraint for the company is 100X+100Y=400000. What is the maximum production?

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is meant by Bio-inspired computing? Write some the popular Bio-inspired computing algorithm. (10)
   b) Why do we prefer heuristic approaches over classical methods to solve optimization problems? Explain with example. (10)
   c) Explain the trade-off criteria for deciding whether to use a heuristic for solving a given problem. (08)
   d) Define Genetic Algorithm (GA). Write simple GA. Write nature to computer mapping. (07)

6. a) Explain exploitation and exploration with examples. (10)
   b) Why do we not use one point crossover to solve TSP? Explain. (07)
   c) State the differences between roulette wheel selection and tournament selection. (08)
   d) Solve traveling salesman problem using particle swarm optimization. (10)

7. a) What is Swarm Intelligence (SI)? What are the properties of SI system? (10)
   b) Write PSO algorithm. What problems are arise when PSO is used in multimodal problems? (10)
   c) Write working procedure of MPSO. (05)
   d) Differentiate between Gbest swarm and Lbest swarm. (10)

8. a) Describe the important of transition probability in ACO algorithm. (10)
   b) "Pheromone evaporation pushes ACO convergence." is it true? Justify your opinion. (07)
   c) Write ACO algorithm. (08)
   d) Explain the employed bee phase, on looker bee phase and scout bee phase in ABC optimization algorithm. (10)